VIEW POINT SYSTEM

MAKING FUTURE REALITY

LIVESTREAM API DOCUMENTATION

Version: v0.1.0

Date: 14. November 2025

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1. Introduction

This document is describing the **API** of the systems from **Viewpointsytem GmbH**. It contains all necessary information to getting started and also includes some examples. For simplification the examples are described using the *curl* command line.

2. Livestream

2.1. Base URL

There are multiple livestream **API** available:

- rtsp://UNIT-IP/api/livestream.whep
- http://UNIT-IP/api/livestream.sdp
- http://UNIT-IP/api/livestream.mp4

This endpoints are following the same rules in regarding of building the pipelines.

2.2. WHEP Protocol

The **WebRTC-HTTP Egress Protocol (WHEP)** defines a simple, **HTTP**-based signaling mechanism that enables a **WebRTC** client to consume unidirectional media streams.

For more information please consult the online documentation[1]

2.3. URL creation

The URL can be build up from the given sources and their elements (defined via keys). It is possible to generate a custom url with this approach.

- Example:

```
?media=mediaSource1
  .keyA(valueA)
  .keyB(
    .subkeyB1(valueB1)
    .subkeyB2(
      .subkeyB2a(valueB2a)
      .subkeyB2b(valueB2b)
    )
  )
&media=mediaSource2
  .keyA(valueA)
  .keyB(valueB)
```

2.4. Media Sources

2.4.1. media=frontcam.codec(CODEC))

Add the front camera element to the **Pipeline** with the codec CODEC.

- CODEC options:
  - h264

2.4.1.1. .gazepoint(.color(RGB))

Add the **Gaze Point** overlay to the **Pipeline**. The **Gaze Point** color can be changed to any RGB value using the RGB key.

2.4.1.2. .heatmap(.timescale(SECONDS))

Add the **Heatmap** overlay to the **Pipeline**. Set the **Heatmap** time to SECONDS.

- SECONDS options:

[1] https://www.ietf.org/archive/id/draft-ietf-wish-whep-03.html

- 1
- 2
- 4
- 8

2.4.2. media=eyecams.codec(CODEC)

Add the **Eye Cameras** to the **Pipeline** with the codec CODEC.

- CODEC options:
  - h264

2.4.3. media=gazedata.format(klv).style(TYPE)

The **Gaze Data** is encrypted with the **KLV** (key-length-values) protocol. Get the **Gaze Data** with **Gaze Data** type TYPE.

- TYPE options:
  - basic
  - advanced

2.4.3.1. Keys

The root **KLV** keys for the can be retrieved from Table 1.

Table 1: Root **KLV** keys

Key	Value
GazeDataKlv	06 0e 2b 34 02 0b 01 01 0f 01 01 01 00 00 00 00
FrontCameraKlv	06 0e 2b 34 02 0b 01 01 0f 01 01 01 00 00 00 01
NineDofKlv	06 0e 2b 34 02 0b 01 01 0f 01 01 01 00 00 00 02
AmbientLightKlv	06 0e 2b 34 02 0b 01 01 0f 01 01 01 00 00 00 03

2.4.3.1.1. Gaze Data KLV

The keys for the **Gaze Data KLV** can be retrieved from Table 2. For more information regarding the **Gaze Data**, please consult the **Gazefile documentation** on the **Unit** under: **WebUI** > **Help** > **Gazefile**

For the explanation of the data type please consult Section 3.

Table 2: **KLV** keys for **Gaze Data**

Key	Value	Data Type	Gazefile Documentation
1	glasses_timestamp	u64	GazeTimeStamp
2	confidence	bool	Confidence
3	gaze_2d	[f64; 2]	Gaze2dX Gaze2dY
4	blink	u8	Blink
5	pupil_diameter_right	f64	PupilDiaRight
6	pupil_diameter_left	f64	PupilDiaLeft
7	gaze_class	u8	GazeClass
8	vergence_angle	f64	VergenceAngle
9	gaze_right_confidence	bool	GazeRightConfidence

10	eyeball_center_right	[f64; 3]	EyeballCenterRightX EyeballCenterRightY EyeballCenterRightZ
11	pupil_center_right	[f64; 3]	PupilCenterRightX PupilCenterRightY PupilCenterRightZ
12	gaze_right	[f64; 3]	GazeRightX GazeRightY GazeRightZ
13	gaze_right_reciprocal_distance	f64	GazeRightReciprocalDistance
14	gaze_right_angular_speed	f64	GazeRightAngularSpeed
15	eye_closed_right	u8	EyeClosedRight
16	pupil_2d_right_confidence	bool	Pupil2dRightConfidence
17	pupil_2d_right	[f64; 5]	Pupil2dRightX Pupil2dRightY Pupil2dRightZ Pupil2dRightA Pupil2dRightB Pupil2dRightO
18	gaze_left_confidence	bool	GazeLeftConfidence
19	eyeball_center_left	[f64; 3]	EyeballCenterLeftX EyeballCenterLeftY EyeballCenterLeftZ
20	pupil_center_left	[f64; 3]	PupilCenterLeftX PupilCenterLeftY PupilCenterLeftZ
21	gaze_left	[f64; 3]	GazeLeftX GazeLeftY GazeLeftZ
22	gaze_left_reciprocal_distance	f64	GazeLeftReciprocalDistance
23	gaze_left_angular_speed	f64	GazeLeftAngularSpeed
24	eye_closed_left	u8	EyeClosedLeft
25	pupil_2d_left_confidence	bool	Pupil2dLeftConfidence
26	pupil_2d_left	[f64; 5]	Pupil2dLeftX Pupil2dLeftY Pupil2dLeftZ Pupil2dLeftA Pupil2dLeftB Pupil2dLeftO

2.4.3.1.2. Front Camera KLV

The keys for the **Front Camera KLV** can be retrieved from Table 3.

Table 3: **KLV** keys for **Front Camera**

Key	Value	Data Type	Gazefile Documentation
1	glasses_timestamp	u64	FrontTimeStamp

2.4.3.1.3. Nine Dof KLV

The keys for the **Nine Dof KLV** can be retrieved from Table 4.

Table 4: **KLV** keys for **Nine Dof**

Key	Value	Data Type	Gazefile Documentation
1	magnetic_field_strength	[f64; 3]	MagneticFluxDensityX MagneticFluxDensityY MagneticFluxDensityZ
2	angular_velocity	[f64; 3]	AngularVelocityX AngularVelocityY AngularVelocityZ
3	orientation	[f64; 4]	OrientationA OrientationB OrientationC OrientationD
4	linear_acceleration	[f64; 3]	LinearAccelerationX LinearAccelerationY LinearAccelerationZ

2.4.3.1.4. Ambient Light KLV

The keys for the **Ambient Light KLV** can be retrieved from Table 5.

Table 5: **KLV** keys for **Ambient Light**

Key	Value	Data Type	Gazefile Documentation
1	light	u16	FrontIlluminance

2.5. Endpoints

2.5.1. GET /api/livestream.mp4

Returns a fragmented mp4 stream with the created URL (see Section 2.3).

- Example: curl "http://UNIT-IP/api/livestream.mp4?media=frontcam.codec(h264)" -o video.mp4

3. Data Types

Table 6: Data Type legend

Abbreviation	Data type	Example
f	Floating point	f64
u	Unsigned integer	u8
bool	Boolean	true
[]	Array	[f64; 3]